Predictors of Waist-to-Height Ratio Among U.S. Adults: A Multiple Linear Regression Analysis Using NHANES 2017–2018 Data

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Background

National health and nutrition demographics provide a basis for exploring factors that influence overall health. Identifying the primary determinants of health and nutrition is essential for uncovering fundamental health trends within a population sample. Data from the National Health and Nutrition Examination Survey (NHANES) can yield useful insights to direct targeted community interventions, particularly for populations at greatest risk.

The National Health and Nutrition Examination Survey collects extensive data on physical health indicators. Among these, the waist-to-height ratio (WHR) has emerged in the literature as a particularly robust outcome variable. Elevated WHR is associated with increased visceral fat in the abdominal region. WHR, calculated as waist circumference divided by height, reflects fat distribution rather than overall body weight. Unlike BMI, which describes lean and fat mass, WHR captures information on central adiposity, the excessive accumulation of visceral fat around internal organs, and cardiometabolic risks more directly across diverse populations. This metric is linked to adverse health outcomes, including cardiovascular disease, diabetes, and metabolic syndrome, which together account for millions of deaths annually. Obesity is still a significant public health concern in the United States, and WHR varies substantially across demographic groups such as sex, race/ethnicity, and age. Identifying predictors of WHR may inform effective screening and intervention approaches.

The NHANES is a nationally representative dataset of U.S. civilian adults. The data is obtained through various methods, including surveys, physical health exams, and consultations. Numerous health data points are collected, but among the most important are smoking status and alcohol consumption habits, as well as uniquely identifiable information such as race, ethnicity, education level, and annual family income, which present a clearer picture of the health trends emerging among U.S. civilian adults. NHANES is collected in standardized Mobile Examination Centers to guarantee high-quality measurements. Its complex sampling design, including oversampling of minority subgroups, makes it well-suited for population-level exploratory inference across demographic strata.

This study uses multiple linear regression on the NHANES 2017-2018 dataset to identify the demographic and clinical factors most strongly associated with WHR in a nationally representative sample of adults. The objective is to direct targeted screening and public health intervention approaches for identified communities.

Research Problem

This analysis addresses the following primary research question: Which demographic, clinical, and lifestyle factors are significantly associated with waist-to-height ratio among U.S. adults during the 2017 NHANES survey period? Secondary questions include: Among significant risk factors, which modifiable factors, such as behavioral variables, remain significant after controlling for non-modifiable characteristics such as sex and race/ethnicity? Additionally, do socioeconomic indicators, including education and income, independently predict WHR beyond clinical variables? Collectively, these questions aim to identify risk factors relevant to behavioral interventions and targeted public health resource allocation. The research questions are organized as follows:

- Which risk factors, both modifiable and non-modifiable, best predict the waist-to-height ratio?
- Which behavioral and lifestyle choices are associated with higher WHR?
- Which race and sex predictors are most significant, and which communities need to be prioritized for intervention?

Model and Method

The analysis included exploratory data analysis, variable selection, and model validation. Distributions of all continuous variables were examined to assess model assumptions and determine the need for transformations. Body mass index (BMI) exhibited mild right skew (Shapiro-Wilk $W = 0.945$, $p < .001$), prompting a log transformation prior to modeling. The log-transformed BMI showed improved symmetry (W = 0.994) and was used in place of the raw BMI.

Stepwise regression using the Akaike Information Criterion (AIC) in both directions was applied to the full model, yielding a substantial set of significant predictors. This procedure permits variable removal at each step. The global F-test confirmed that the final model explains significantly more variation in WHR than the null intercept-only model (F = 2957, df = 18, p < 2.2e-16), leading to the rejection of the null hypothesis $H_0 : \beta_1 = \beta_2 = \cdots = \beta_{18} = 0$. For the CAP variable, a nested F-test was used to evaluate its marginal contribution, failing to reject $H_0 : \beta_{CAP} = 0$ (F = 0.63, p = 0.428), which supports its exclusion. ALT was similarly excluded via nested F-test (F = 2.73, p = 0.099), since its marginal significance was not significant at the 0.05 level.

The sample comprised approximately 4,382 civilian adults from the NHANES 2017-2018 cycle, balanced by sex. Participants ranged in age from 18 to 80 years, providing broad representation over the adult lifespan. Race and ethnicity were not equally distributed, with non-Hispanic White participants forming the largest group, followed by Hispanic, non-Hispanic Black, and non-Hispanic Other. Intentional oversampling of minority populations was implemented to ensure adequate statistical power for subgroup analyses.

Under the Gauss-Markov assumptions of linearity, exogeneity, homoscedasticity, and uncorrelated errors, the ordinary least squares (OLS) estimator is the Best Linear Unbiased Estimator, ensuring minimum variance among all linear unbiased estimators. This guarantees that among all estimators that are linear function of the observed data and unbiased, OLS produces the most precise estimates. The estimated residual variance ($\hat{\sigma}^2$) yielded a residual standard error of 0.02876 on 4,363 degrees of freedom. Variance inflation factors were below 10 for all predictors, with waist (VIF = 9.05) and log(BMI)(VIF = 8.53) the highest, indicating no critical multicollinearity. For exploratory purposes, removing these variables from the model considerably reduces the adjusted R-squared; they were kept, but this is a limitation worth noting. Using Mallows' C_p and adjusted R^2 confirmed that all 14 predictors should be retained, as both criteria were minimized with the full set of predictors (Figure 1). Three influential observations (3459, 4712, 2399) were identified using Cook's distance. These participants describe a large residual but low leverage, indicating they are outliers in the response space rather than the predictor, plausibly extreme values. A two-way analysis of variance (ANOVA) was conducted to assess whether mean WHR differed significantly by sex and race/ethnicity, and to evaluate the interaction among these factors. Tukey HSD post-hoc comparisons were subsequently performed to identify specific group pairs with significant differences.

Four assumptions underlie the OLS model: linearity, zero-mean errors, homoscedasticity, and uncorrelated errors. The residuals vs Fitted plot (Figure 6) showed no systematic curvature, supporting linearity. The Q-Q plot revealed heavy tails, particularly at the lower end, indicating modest departures from normality. With the sample size sufficiently large (~4,300), this is unlikely to systematically affect inference under the central limit theorem. For the two-way ANOVA, homogeneity of variance across the eight sex x race/ethnicity cells is assumed; the boxplots in Figure 5 suggest somewhat wider spread among NH-Black males, indicating care in interpreting cross-group comparisons.

Conclusion and Discussion

This analysis found that a combination of demographic, socioeconomic, and lifestyle habits can explain over 90% of the variation in waist-to-height ratio (WHR) among U.S. adults. Waist circumference and BMI emerged as the most impactful predictors, which is consistent with their direct role in determining WHR. Their largest t-statistics (52.90 and 23.21, respectively) indicate strong independent contributions, even after controlling for all other predictors. Despite elevated VIFs, both remain essential to the final models' fit. Male participants exhibited significantly lower WHR than females (β = -0.040, p < 0.001), the largest demographic effect in the model, aside from race/ethnicity predictors.

WHR decreased significantly at higher attainment levels (levels 3-5) relative to less than 9th grade, with college graduates showing the largest reduction (β = -0.010, p < 0.001); the 9th-11th grade level did not reach significance (p = 0.252). This suggests the protective effect strengthens at higher levels of educational attainment. Notably, the results also identify that educational attainment serves as a proxy for health-promoting behaviors and access to resources. Higher income was also independently associated with lower WHR (β = -0.0004, p = 0.004), indicating economic resources facilitate healthier dietary patterns. Furthermore, among modifiable behavioral factors, both drinking groups showed significantly lower WHR than non-drinkers. Lastly, greater physical activity was associated with lower WHR (β -0.001, p = 0.004), controlling for all other variables.

Tukey HSD comparison revealed that NH-Other participants differed significantly from all other groups. Together, these findings identify waist circumference, BMI, sex, and race/ethnicity as the primary non-modifiable predictors of WHR, while education, income, alcohol consumption, and physical activity represent modifiable targets for intervention.

Limitations: The cross-sectional NHANES design prevents causal inference. Stepwise selection inflates familywise Type I error, and all findings are exploratory. VIFS were below 10, with waist (9.05) and log(BMI) (8.53) the highest, indicating no critical multicollinearity. Physical activity could not be log-transformed due to zero values, and the residual skew of this predictor could affect inference for that coefficient. Additionally, NHANES uses a stratified probability sampling design, which may induce correlated errors and violate the independence assumption for both regression and ANOVA; survey-weighted methods were not applied, which represents a limitation.

The OLS framework assumes errors are uncorrelated and homoscedastic. The Scale-Location plot revealed a mild upward trend at higher fitted values, suggesting slight heteroscedasticity behavior at the upper end of the WHR distribution. Although it is not severe enough to invalidate inference, it warrants acknowledgment.

Additionally, the hat matrix $H = X(X^\top X)^{-1}X^\top$ was used in Cook's distance calculations; fitted values $\hat{y} = Hy$ represent the unique projection of y onto the column space of X. The three flagged observations have large residuals $(I - H)y$ despite low leverage, confirming they are outliers in the response space rather than the predictor space.

The two-way ANOVA confirmed significant main effects of both sex (F = 81.37, p $<$ .001) and race/ethnicity (F = 41.50, p $<$ .001) on WHR. The interaction term was significant (F = 17.67, p $<$ .001), indicating that sex differences in WHR do vary across racial/ethnic groups. The significant interaction suggests the sex gap in WHR varies in magnitude across racial/ethnic groups, as visible in Figure 5.

Future work should include interaction terms between sex and race/ethnicity, employ long-term studies to determine causality, and conduct analyses stratified by race/ethnicity or age group to further these outcomes and inform community-specific interventions.

Tables and Figures

Table 1: Summary Statistics for Continuous Variables

Variable	Mean	SD	Min	Max
whr	0.60	0.10	0.04	1.03
waist	100.25	17.23	56.40	169.50
log_BMI	3.36	0.23	2.69	4.22
age	49.39	18.18	18.00	80.00
log_triglcd	4.79	0.56	3.22	7.40
bp.di.m	72.87	11.72	8.00	124.67
bp.sy.m	125.87	19.30	78.67	234.00
income	7.49	3.25	1.00	12.00
phy.act	1.08	1.71	0.00	13.74

Table 2

Table 2: Final Model Coefficients

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	-0.23648	0.01293	-18.28460	0.00000
waist	0.00401	0.00008	52.89605	0.00000
sexMale	-0.03992	0.00099	-40.52263	0.00000
log_BMI	0.12786	0.00551	23.20988	0.00000
age	0.00045	0.00003	14.54194	0.00000
race_ethNH-Black	-0.00466	0.00124	-3.75204	0.00018
race_ethHispanic	0.01106	0.00127	8.71361	0.00000
race_ethNH-Other	0.00735	0.00132	5.57338	0.00000
educ_lev2	-0.00246	0.00215	-1.14624	0.25176
educ_lev3	-0.00640	0.00197	-3.25132	0.00116
educ_lev4	-0.00893	0.00195	-4.58078	0.00000
educ_lev5	-0.01016	0.00208	-4.89076	0.00000
hvy.drk1	-0.00428	0.00095	-4.48576	0.00001
hvy.drk2	-0.00593	0.00194	-3.06352	0.00220
bp.di.m	-0.00024	0.00004	-5.47543	0.00000
log_triglcd	0.00348	0.00087	3.99717	0.00007
bp.sy.m	0.00011	0.00003	3.79492	0.00015
income	-0.00043	0.00015	-2.90766	0.00366
phy.act	-0.00077	0.00027	-2.85096	0.00438

Table 3: Variance Inflation Factors

Predictor	VIF
waist	9.052
sex	1.286
log_BMI	8.526
age	1.712
race_eth	1.570
educ_lev	1.522
hvy.drk	1.143
bp.di.m	1.393

log_triglcd	1.264
bp.sy.m	1.797
income	1.195
phy.act	1.128

Table 4: Two-Way ANOVA: WHR by Sex and Race/Ethnicity

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
sex	1	0.8364614	0.8364614	81.37435	0
race_eth	3	1.2798759	0.4266253	41.50384	0
sex:race_eth	3	0.5448740	0.1816247	17.66918	0
Residuals	4374	44.9611218	0.0102792	NA	NA

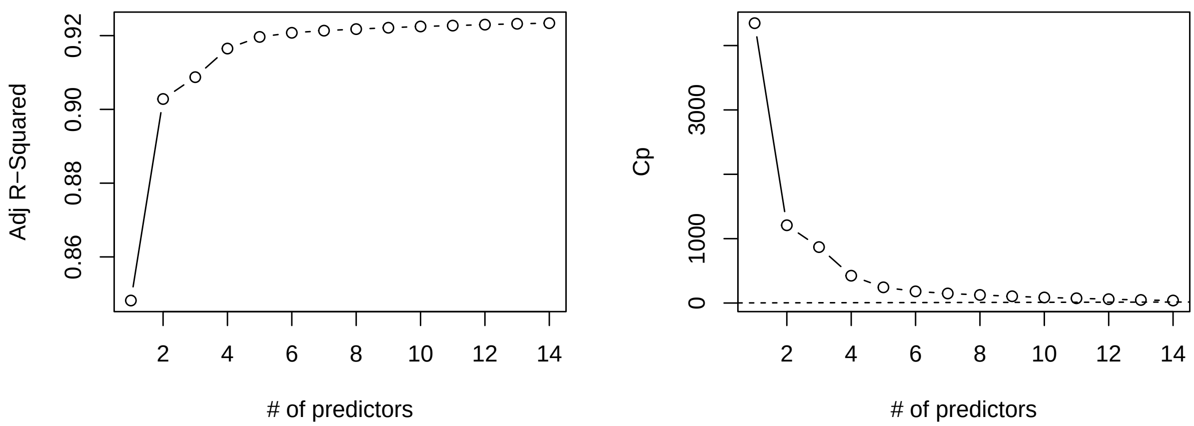

Figure 1: Best subsets regression: adjusted R-squared and Mallows Cp by number of predictors

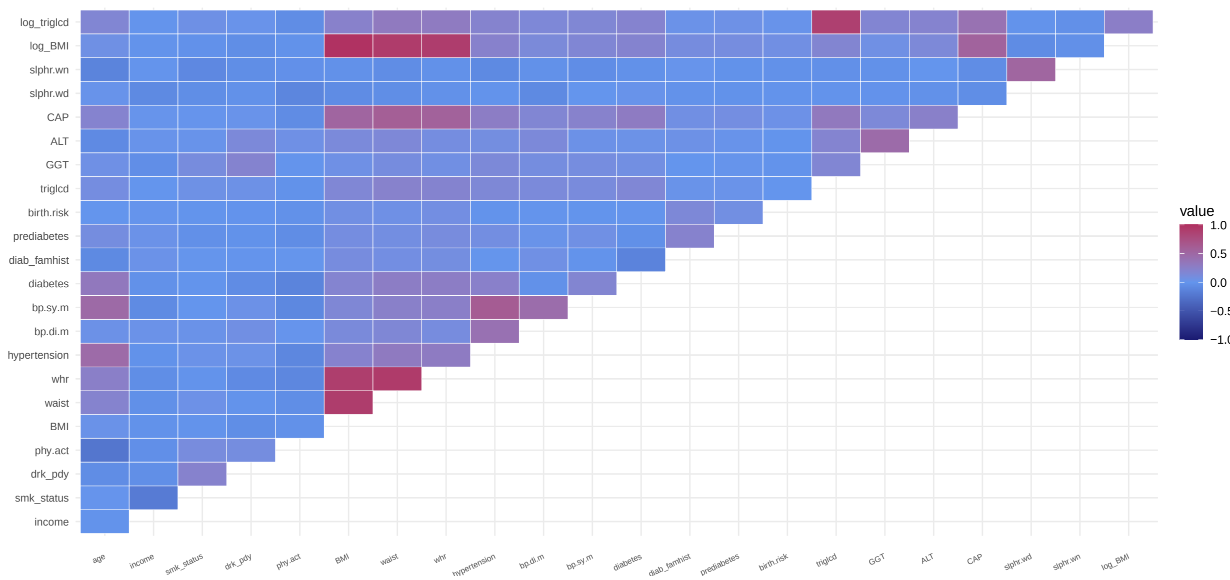

Figure 2: Correlation Matrix

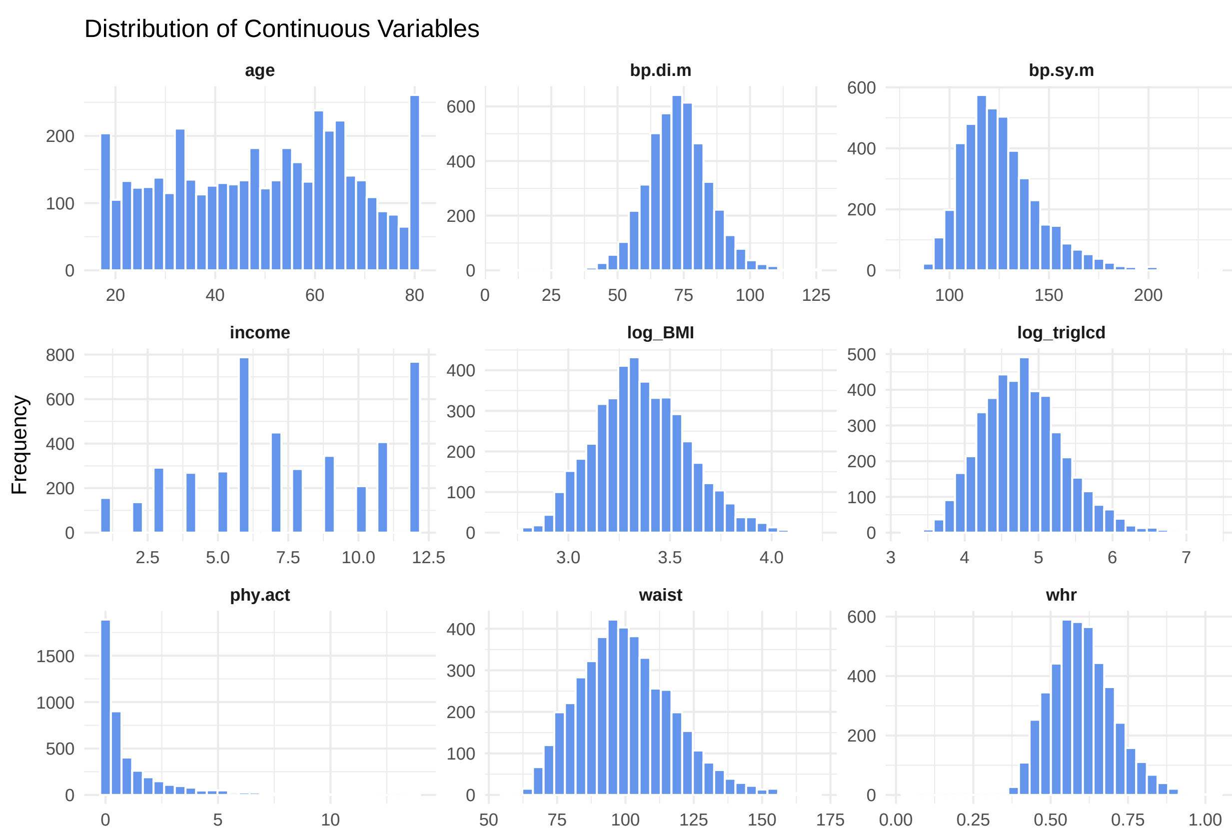

Figure 3: Distribution of key Continuous variables

Figure 4: Distribution of key categorical variables

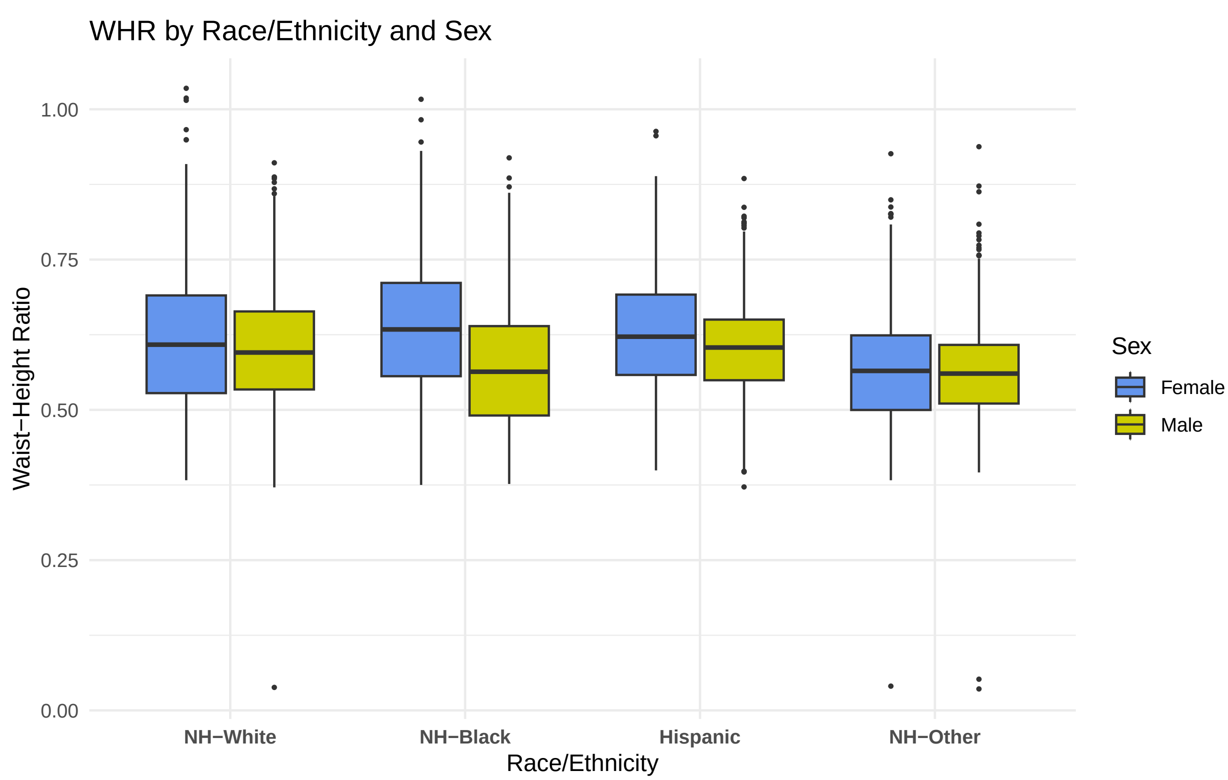

Figure 5: WHR by race/ethnicity and sex (unadjusted)

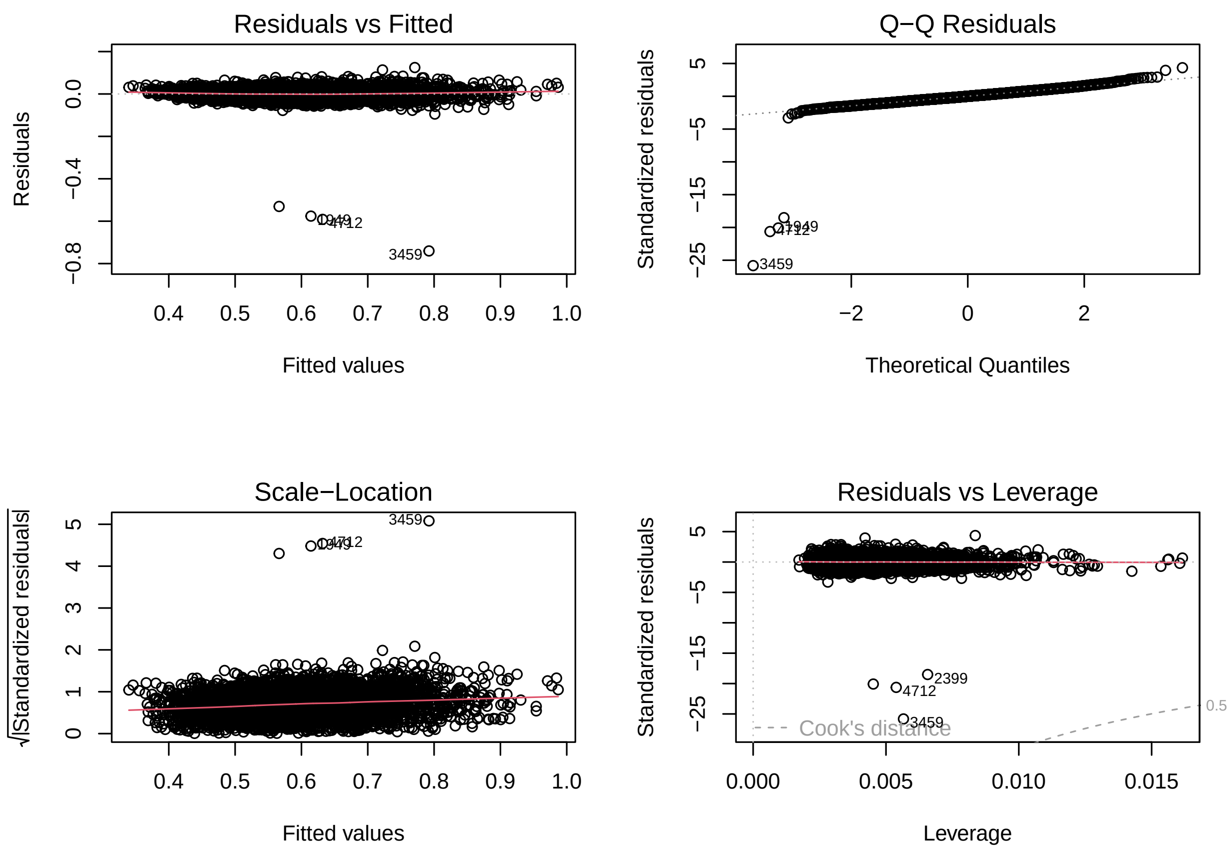

Figure 6: Regression diagnostic plots for final model

Appendix: R Code

The following code reproduces all analysis presented in this report. The Code is organized as follows: data preparation, exploratory data analysis, model fitting, model validation, and visualization.

```
knitr::opts_chunk$set(echo = FALSE, warning = FALSE, message = FALSE)


library(tidyverse)
library(reshape2)
library(knitr)
library(kableExtra)
library(car)
library(leaps)

load("nhanes17.RData")

# Transformations
nhanes17$log_BMI <- log(nhanes17$BMI)
nhanes17$log_triglcd <- log(nhanes17$triglcd)

nhanes17$hvy.drk <- as.factor(nhanes17$hvy.drk)
nhanes17$educ_lev <- as.factor(nhanes17$educ_lev)

# Factor variables

nhanes17$hvy.drk <- as.factor(nhanes17$hvy.drk)
nhanes17$educ_lev <- as.factor(nhanes17$educ_lev)

# Correlation plot
nhnames_num <- nhanes17[, sapply(nhanes17, is.numeric)]
nhnames_num <- nhnames_num[, !colnames(nhnames_num) %in% "ID"]

# Correlation matrix
nh_cor <- nhanes17 %>%
  select(whr, waist, log_BMI, age, log_triglcd, bp.di.m,)
nhnames_num <- nhanes17[, sapply(nhanes17, is.numeric)]
nhnames_num <- nhnames_num[, !colnames(nhnames_num) %in% "ID"]
nh_cor      <- cor(nhnames_num, use = "pairwise.complete.obs")
cor_melt    <- melt(nh_cor)
cor_melt <- cor_melt[as.integer(cor_melt$Var1) < as.integer(cor_melt$Var2), ]
# setup chunk - transformations
nhanes17$log_BMI     <- log(nhanes17$BMI)
nhanes17$log_triglcd <- log(nhanes17$triglcd)

# factor conversions
nhanes17$hvy.drk  <- as.factor(nhanes17$hvy.drk)
nhanes17$educ_lev <- as.factor(nhanes17$educ_lev)
nhanes17$sex      <- as.factor(nhanes17$sex)
nhanes17$race_eth <- as.factor(nhanes17$race_eth)

# Model Building

full_model <- lm(whr ~ . - ID -BMI -triglcd, data = nhanes17)
```

```r
null_model <- lm(whr ~ 1,      data = nhanes17)
mod_step   <- step(null_model, direction = "both",
                   scope = formula(full_model))
summary(mod_step)

# Final Model - ALT droppped using nested F-test
final_mod <- lm(formula = whr ~ waist + sex +
                  log_BMI + age + race_eth + educ_lev +
                  hvy.drk + bp.di.m +
                  log_triglcd + bp.sy.m + income + phy.act,
                data = nhanes17)
summary(final_mod)


final_mod_no_bmi <- lm(formula = whr ~  sex +
                    age + race_eth + educ_lev +
                  hvy.drk + bp.di.m +
                  log_triglcd + bp.sy.m + income + phy.act,
                data = nhanes17)
summary(final_mod_no_bmi)



# Nested F-test: ALT
anova(final_mod, mod_step)

# Nested F-test: CAP
mod_cap <- lm(whr ~ waist + sex + log_BMI + age + race_eth +
                educ_lev + hvy.drk + bp.di.m + log_triglcd +
                bp.sy.m + income + phy.act + CAP,
              data = nhanes17)
anova(final_mod, mod_cap)

# VIF
vif(final_mod)

# Best subsets (uses log_triglcd - consistent with final model)
best_sub     <- regsubsets(whr ~ waist + sex + race_eth + log_BMI + age +educ_lev + hvy.drk + log_trigl
                             bp.sy.m + phy.act + income,
                           data = nhanes17, nvmax = 14)
summary_best <- summary(best_sub)
which.max(summary_best$adjr2)
which.min(summary_best$cp)

# Two-way ANOVA
anova_model <- aov(whr ~ sex * race_eth, data = nhanes17)
summary(anova_model)
tukey   <- TukeyHSD(anova_model, which = "race_eth")
tukey
# BMI is skewed
shapiro.test(nhanes17$BMI)
shapiro.test(nhanes17$log_BMI)
```

```r
shapiro.test(nhanes17$triglcd)
shapiro.test(nhanes17$log_triglcd)
shapiro.test(nhanes17$phy.act)
shapiro.test(nhanes17$log_triglcd)

par(mfrow = c(1,2))
hist(nhanes17$BMI)
hist(nhanes17$log_BMI)

shapiro.test(nhanes17$log_BMI)
library(leaps)
best_sub <- regsubsets(whr ~ waist + sex + race_eth + log_BMI + age +
                         educ_lev + hvy.drk + triglcd + bp.di.m +
                         bp.sy.m + phy.act + income,
                       data = nhanes17, nvmax = 14)

summary_best <- summary(best_sub)

# which model is best by each criterion
which.max(summary_best$adjr2)
which.min(summary_best$cp)
nhanes17 %>%
  select(whr, waist, log_BMI, age, log_triglcd, bp.di.m, bp.sy.m, income, phy.act) %>%
    summarise(across(everything(), list(
      Mean = ~round(mean(., na.rm = TRUE), 2),
      SD = ~round(sd(., na.rm = TRUE), 2),
      Min = ~round(min(., na.rm = TRUE), 2),
      Max = ~round(max(., na.rm = TRUE), 2)
      ))) %>%
      pivot_longer(everything(),
                   names_to = c("Variable", ".value"),
                   names_pattern  = "^(.+)_(Mean|SD|Min|Max)$") %>%
      kable(caption = "Summary Statistics for Continuous Variables",
            booktabs = TRUE) %>%
      kable_styling(latex_options = c("striped", "hold_position"), font_size = 9)
summary(final_mod)$coefficients %>%
  as.data.frame() %>%
  round(5) %>%
  kable(caption = "Final Model Coefficients", booktabs = TRUE) %>%
  kable_styling(latex_options = c("striped", "hold_position"), font_size = 9)
vif(final_mod) %>%
  as.data.frame() %>%
  rownames_to_column("Predictor") %>%
  select(Predictor, GVIF) %>%
  rename(VIF = GVIF) %>%
  mutate(VIF = round(VIF, 3)) %>%
  kable(caption = "Variance Inflation Factors", booktabs = TRUE) %>%
  kable_styling(latex_options = c("striped", "hold_position"), font_size = 9)
anova_model <- aov(whr ~ sex * race_eth, data = nhanes17)

summary(anova_model)[[1]] %>%
  as.data.frame() %>%
  kable(caption = "Two-Way ANOVA: WHR by Sex and Race/Ethnicity", booktabs= TRUE) %>%
```

```r
  kable_styling(latex_options = c("striped", "hold_position"), font_size = 9)

par(mfrow = c(1, 2))
plot(summary_best$adjr2, type = "b",
     xlab = "# of predictors", ylab = "Adj R-Squared")
plot(summary_best$cp, type = "b",
     xlab = "# of predictors", ylab  = "Cp")
abline(a = 1, b = 1, lty = 2)
ggplot(cor_melt, aes(Var1, Var2, fill = value))+
  geom_tile(color = "white")+
  scale_fill_gradient2(low = "midnightblue", mid = "cornflowerblue", high = "maroon",
                       midpoint = 0, limits = c(-1, 1))+
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 25, hjust = 1, vjust = 0.5, size = 6),
        axis.text.y = element_text(size = 8))+
  labs(x = NULL, y = NULL)
nhanes17 %>%
  select(whr, waist, log_BMI, age, log_triglcd, bp.di.m, bp.sy.m, income, phy.act) %>%
  pivot_longer(everything(),
               names_to = "variable",
               values_to = "value") %>%
  ggplot(aes(x = value))+
  geom_histogram(bins = 30, fill = "cornflowerblue", color = "white")+
  facet_wrap(~variable, scales = "free")+
  theme_minimal()+ theme(strip.text = element_text(face = "bold"))+
  labs(title = "Distribution of Continuous Variables",
       x = NULL, y = "Frequency")
# select your categorical predictors
nhanes17 |>
  select(sex, race_eth, educ_lev, hvy.drk) |>
  mutate(across(everything(), as.factor)) |>
  pivot_longer(everything(), names_to = "variable", values_to = "value") |>
  ggplot(aes(x = value, fill = variable)) +
  geom_bar(alpha = 0.85, color = "white") +
  facet_wrap(~variable, scales = "free") +
  scale_fill_brewer(palette = "Dark2") +
  theme_minimal() +
  theme(legend.position = "none",
        strip.text = element_text(face = "bold"),
        axis.text.x = element_text(angle = 20, hjust = 1)) +
  labs(title = "Distributions of Categorical Variables", x = NULL, y = "Count")
nhanes17 %>%
  ggplot(aes(x = race_eth, y = whr, fill = sex)) +
  geom_boxplot(outlier.size = 0.5) +
  scale_fill_manual(values = c("cornflowerblue", "yellow3"),
                    labels = c("Female", "Male")) +
  theme_minimal() +
  labs(title = "WHR by Race/Ethnicity and Sex",
       x = "Race/Ethnicity", y = "Waist-Height Ratio",
       fill = "Sex") +
  theme(axis.text.x = element_text(face = "bold"))
par(mfrow = c(2, 2))
plot(final_mod)
```